Edition 13.07.2017

This is a description of the code SOLPS-ITER output for key ‘B2wdat’ ‘4’ defined in the file b2mn.dat. The first part of the name of almost each file gives the procedure from which it was taken.

The explicit expression for different variables not described below in details can be sought in the document B2solps5.2_equations_24.04.2017.doc and its later versions. Only new expressions are given for each term in this document. Alternative older expressions are given in B2solps5.2_equations_24.04.2017.doc and its later versions.

GEOMETRY

…from b2wdat

$\sqrt{g}$ - ***vol*** cell volume; h_x, h_y, h_z **- *hx,hy,hz*** cell dimensions; $\frac{\sqrt{g}}{h_x} b_x$ - ***pbsx***

$h_y \cos\theta$ **- *hy1,*** where θ is the complementary angle between the x-direction and the y-direction on the (ix,iy) cell.

B_x, B_z **- *bbx, bbz*** ; B **- *bb*** ;

crx0..3, cry0..3 – physical Cartesian coordinates (y axis coincides with the symmetry axis of tokamak) of the corners of the mesh. Correspondingly: left lower corner, right lower corner, right upper corner, left upper corner of the cell in the mesh transformed to the orthogonal one.

rightix, rightiy, leftix, leftiy, topix, topiy, bottomix, bottomiy matrices containing indexes of cells which are neighboring to the chosen cell at the physical mesh, including cuts.

Flows are given at the left and lower faces of the cell (meaning left and lower in the orthogonal mesh representation). Source terms are given in the centers of the cell.

CONTINUITY EQUATION

$$\frac{\partial n_a}{\partial t} + \frac{1}{\sqrt{g}}\frac{\partial}{\partial x}\left(\frac{\sqrt{g}}{h_x}\Gamma_{ax}\right) + \frac{1}{\sqrt{g}}\frac{\partial}{\partial y}\left(\frac{\sqrt{g}}{h_y}\Gamma_{ay}\right) = S_a^n, \qquad (1)$$

a = 0, 1,…,n$_s$*-1*- indexes of particles

Here and further, numeration of sorts of particles: first main particles starting from neutrals(000) to the highest charged state, then impurities of the first sort, starting from neutrals and to the highest charged state, then impurity of the second sort and so on.

The residuals of the iteration of this equation are ***b2npc9_resco001… ns-1*** for ions and ***b2npco_resco000*** for neutrals.

n_a - ***b2npc9_na001… ns-1*** for ions and ***b2npco_na000***… for neutrals

$-\sqrt{g}\frac{\partial n_a}{\partial t}$ - ***b2srdt_snadt001… ns-1***

$S_a^n\sqrt{g}$ - ***b2npc9_sna001… ns-1*** for ions and ***b2npco_sna000…*** for neutrals.

$$S_a^n = S_{I_a}^n + S_{I_{a+1}}^n + S_{R_a}^n + S_{R_{a-1}}^n + S_{CX_a}^n + S_{CX_{a-1}}^n (+S_a^{eirene})(+S_a^{BC})$$

For source terms here and further either Monte-Carlo code EIRENE or fluid neutral modeling can be used. In case if EIRENE is used, the neutral density in all sources using fluid neutrals is multiplied on small scaling factor. Therefore these sources are not exactly zero but they become negligibly small.

Different particle source components:

$(S^n_{I_a} + S^n_{I_{a+1}})\sqrt{g}$ - ***b2stel_sna_ion001… ns-1*** ion source due to ionization of ions and in case of fluid neutrals - of neutrals;

$(S^n_{R_a} + S^n_{R_{a-1}})\sqrt{g}$ - ***b2stel_sna_rec001… ns-1*** ion source due to recombination to lower ionization states of ions and in case of fluid neutrals - to neutrals;

$(S^n_{CX_a} + S^n_{CX_{a-1}})\sqrt{g}$ - ***b2stcx_sna_ion001… ns-1*** ion source due to charge-exchange with ions and in case of fluid neutrals - with neutrals.

$S_a^{eirene}\sqrt{g}$ - ***b2stbr_sna_eir001…ns-1*** ion source due to recombination to and ionization of neutrals calculated in Monte-Carlo code EIRENE.

$S_a^{BC}\sqrt{g}$ - ***b2stbc_phys_sna001…ns-1*** ion source in the guard cells due to boundary conditions. The flux to the guard cells, defined in boundary condition is implemented as a sink in these cells.

$\frac{\sqrt{g}}{h_x}\tilde{\Gamma}_{ax}$ - ***b2npc9_fnax001…ns-1*** for ions and ***b2npco_fnax000*** for neutrals

$\frac{\sqrt{g}}{h_x}\tilde{\Gamma}_{ay}$ - ***b2npc9_fnay001…ns-1*** for ions and ***b2npco_fnay000*** for neutrals

For ions that is effective flow, not a physical one, but it gives the same divergence as a physical one.

$$\tilde{\Gamma}_{ax} = b_x V_{\parallel a} n_a + V_{ax}^{(E)} n_a + V_a^{corr\_dpc} n_a - D_a^n \frac{1}{h_x}\frac{\partial n_a}{\partial x} - \tilde{D}_{ax}^p \frac{1}{h_x}\frac{\partial p_a}{\partial x} - D_{axy}\frac{1}{h_y}\frac{\partial n_a T_i}{\partial y}$$

$$\tilde{\Gamma}_{ay} = \begin{cases} V_{ay}^{(E)} n_a + V_{ay}^{(AN)} n_a + \dfrac{j_y^{(AN)} + j_y^{(in)} + \tilde{j}_y^{(vis\parallel)}}{e} n_a - D_a^n \dfrac{1}{h_y}\dfrac{\partial n_a}{\partial y} - \tilde{D}_{ay}^p \dfrac{1}{h_y}\dfrac{\partial p_a}{\partial y} + D_{ayx}\dfrac{1}{h_x}\dfrac{\partial n_a T_i}{\partial x} & a = 1 \\ V_{ay}^{(E)} n_a + V_{ay}^{(AN)} n_a - D_a^n \dfrac{1}{h_y}\dfrac{\partial n_a}{\partial y} - \tilde{D}_{ay}^p \dfrac{1}{h_y}\dfrac{\partial p_a}{\partial y} + D_{ayx}\dfrac{1}{h_x}\dfrac{\partial n_a T_i}{\partial x} & a \neq 1 \end{cases}$$

$\frac{\sqrt{g}}{h_x} b_x V_{\parallel a} n_a$ - parallel flow ***b2tfnb_bxuanax001…ns-1*** for ions and ***b2tfnb_bxuanax000*** for neutrals

$\frac{\sqrt{g}}{h_x} V_{ax,y}^{(E)} n_a$ - ExB flows ***b2tfnb_vaecrbnax,y001…ns-1*** for ions. For neutrals these fluxes are ***0***.

$\frac{\sqrt{g}}{h_x} V_a^{corr-dpc} n_a$ - numerical correction ***b2tfnb_dpccornax001…ns-1*** for ions and ***b2tfnb_dpccornax00*** for neutrals

$\frac{\sqrt{g}}{h_x} V_{ay}^{(AN)} n_a$ - anomalous pinch flow ***b2tfnb_cvlbnay001…ns-1*** for ions. For neutrals these fluxes are ***0***.

$-D_a^n \frac{\sqrt{g}}{h_x}\frac{1}{h_x}\frac{\partial n_a}{\partial x}$, $-D_a^n \frac{\sqrt{g}}{h_y}\frac{1}{h_y}\frac{\partial n_a}{\partial y}$ - anomalous diffusion ***b2tfnb_dPat_mdf_gradna{x,y}001…ns-1*** for ions. These are the values which actually enter the particle balance in the code. For neutrals these fluxes are ***0***.

Diffusion coefficients are modified for the numerical stability according to Patankar's scheme:

$$D_{x,y}^{\mathrm{modified}} = \sqrt{D_{x,y}^2 + (V_{x,y}^{convective} h_{x,y}/2)^2} \ .$$

$D_a^n \frac{\sqrt{g}}{h_{x,y}^{\ 2}}$ - ***b2trno_cdnax,y001…ns-1*** ion anomalous diffusion coefficients before the Patankar correction.

D_a^n - ***b2tqna_dna001…ns-1*** (corresponding to cell centers)

$-\tilde{D}_{ax}^p \frac{\sqrt{g}}{h_x}\frac{1}{h_x}\frac{\partial p_a}{\partial x}, -\tilde{D}_{ax}^p \frac{\sqrt{g}}{h_x}\frac{1}{h_x}\frac{\partial p_a}{\partial x}$ - ***b2tfnb_dgradpbx,y000*** for neutrals. For ions these fluxes are ***0***.

$\tilde{D}_{ax,y}^p \frac{\sqrt{g}}{h_{x,y}^{\ 2}}$ - ***b2trno_cdpax,y*** neutral diffusion coefficients before the Patankar correction but after flux-limiting.

$$\tilde{D}_{ax,y}^p = \begin{cases} \dfrac{D_a^p}{\left(1+\left(\dfrac{\frac{\sqrt{g}}{h_{x,y}} D_a^p \left|\frac{1}{h_{x,y}}\frac{\partial p_a}{\partial x,y}\right|}{\alpha \frac{\sqrt{g}}{h_{x,y}} n_a \frac{V_{T_i}}{4}}\right)^{\gamma_1}\right)^{\frac{1}{\gamma_1}}}, \quad V_{T_i} = \sqrt{\dfrac{8T_i}{\pi m_a}}, \quad z_a = 0 \\ D_a^p, \qquad z_a \neq 0 \end{cases}$$

D_a^p - ***b2tqna_dpa000…ns-1***

$D_{axy} \frac{\sqrt{g}}{h_x}\frac{1}{h_y}\frac{\partial n_a T_i}{\partial y}, \quad -D_{ayx}\frac{\sqrt{g}}{h_y}\frac{1}{h_x}\frac{\partial n_a T_i}{\partial x}$ - ***b2tfnb_fnbPSch{x,y}001…ns-1*** for ions and ***0*** for neutrals.

Mind the sign! It is opposite in output file and in flux balance. These are effective fluxes, giving the same divergence as diamagnetic ones.

$$\begin{cases} D_{axy} = D_{ayx} = \left(\dfrac{1}{B^2} - \dfrac{1}{\langle B^2\rangle}\right)\dfrac{B_z}{z_a e}, & z_a \neq 0 \\ D_{axy} = 0, & z_a = 0 \end{cases}, \text{ where } \langle B^2 \rangle = \frac{\iint_{core} B^2 \sqrt{g}\,dxdy}{\iint_{core}\sqrt{g}\,dxdy}$$

Velocities:

Correction $$V_a^{corr\_dpc} = \frac{c^{corr\_dpc}}{16}\frac{B_x}{B}\sqrt{\frac{p_a}{m_a n_a}}\frac{\partial^2}{\partial x^2}\left(\frac{1}{p_a}\frac{\partial p_a}{\partial x}\right)$$

E×B drift

$$V_{ax}^{(E)} = \begin{cases} -c_{E\times B}\dfrac{B_z}{B^2}\dfrac{1}{h_y}\dfrac{\partial \phi}{\partial y}, & z_a \neq 0, \\ 0, & z_a = 0 \end{cases}, \qquad V_{ay}^{(E)} = \begin{cases} c_{E\times B}\dfrac{B_z}{B^2}\dfrac{1}{h_x}\dfrac{\partial \phi}{\partial x}, & z_a \neq 0, \\ 0, & z_a = 0 \end{cases},$$

∇B drift

$$\tilde{V}_{xa}^{(dia)} = \begin{cases} c_{dia}\dfrac{T_i B_z}{Z_a e}\dfrac{\partial}{h_y \partial y}\left(\dfrac{1}{B^2}\right), & z_a \neq 0, \\ 0, & z_a = 0 \end{cases}, \qquad \tilde{V}_{ya}^{(dia)} = \begin{cases} -c_{dia}\dfrac{T_i B_z}{Z_a e}\dfrac{\partial}{h_x \partial x}\left(\dfrac{1}{B^2}\right), & z_a \neq 0, \\ 0, & z_a = 0 \end{cases},$$

Diamagnetic drift

$$W_{a\perp}^{(dia)} = \begin{cases} -\dfrac{B_z}{Z_a e B^2 n_a}\dfrac{\partial n_a T_i}{h_y \partial y}, & z_a \neq 0, \\ 0, & z_a = 0 \end{cases}, \qquad W_{ay}^{(dia)} = \begin{cases} \dfrac{B_z}{Z_a e B^2 n_a}\dfrac{\partial n_a T_i}{h_x \partial x}, & z_a \neq 0, \\ 0, & z_a = 0 \end{cases}$$

$V_{x,y}^{(E)}$ - ***b2tfnb_vbecrbx,y001***

$\tilde{V}_{x,y}^{(dia)}$ - ***b2tfnb_vbdiax,y001…ns-1***

$W_{x,y}^{(dia)}$ - ***b2tfnb_wbdiax,y001…ns-1***

$V_{ay}^{(AN)}$ - ***b2tqna_vla0y000…ns-1***

$\dfrac{\sqrt{g}}{h_x}\tilde{V}_{ax,y}^{(dia)} n_a$ - ***b2tfnb_vadianax,y001…ns-1*** for ions and ***0*** for neutrals

$n_e = \sum_{a=0}^{n_s-1} z_a n_a$ - ***ne***,

Kinetic energy

$$E_{k,a} = \frac{m_a}{2}\left[u^2{}_{\|a} + \left(W_{a\perp}^{(dia)} + V_{a\perp}^{(ExB)}\right)^2 + \left(W_{ay}^{(dia)} + V_{ay}^{(ExB)}\right)^2\right]$$ - ***b2tfnb_kbnrgy001…ns-1***

MOMENTUM BALANCE

$$m_a \frac{\partial n_a V_{\|a}}{\partial t} + \frac{1}{h_z\sqrt{g}}\frac{\partial}{\partial x}\left(\frac{h_z\sqrt{g}}{h_x}\Gamma_{ax}^m\right) + \frac{1}{h_z\sqrt{g}}\frac{\partial}{\partial y}\left(\frac{h_z\sqrt{g}}{h_y}\Gamma_{ay}^m\right) + \frac{b_x}{h_x}\frac{\partial n_a T_i}{\partial x} + Z_a e n_a \frac{b_x}{h_x}\frac{\partial \varphi}{\partial x} = \qquad (2)$$

$$= S_{a\|}^m + S_{CF_a}^m + S_{fr_a}^m + S_{Term\,a}^m + S_{I_a}^m + S_{R_a}^m + S_{CX_a}^m$$

$V_{\|a}$ - ***b2npmo_ua000…ns-1***

The residuals of the iteration of this equation are ***b2npmo_resmo000…ns-1*** .

$S_a^m h_z\sqrt{g}$ - ***b2npmo_smb000…ns-1*** : full r.h.s. of the equation

$-m_a \dfrac{\partial n_a V_{\|a}}{\partial t} h_z\sqrt{g}$ - ***b2srdt_smodt000…ns-1***

$\dfrac{h_z\sqrt{g}}{h_x}\Gamma^m{}_{ax,y}$ - ***b2nmo_fmox,y000…ns-1***

$S^m_{CF} h_z \sqrt{g} = m_a b_x n_a V^2_{\|a} \frac{1}{h_z h_x} \frac{\partial h_z}{\partial x} h_z \sqrt{g}$ - ***b2npmo_smocf000…ns-1*** centrifugal force

$-\frac{b_x}{h_x} \frac{\partial n_a T_i}{\partial x} h_z \sqrt{g}$ **- *b2sigp_smogpi000…ns-1***

$-Z_a e n_a \frac{b_x}{h_x} \frac{\partial \varphi}{\partial x} h_z \sqrt{g}$ - ***b2sigp_smogpo000…ns-1***

$\left(S^m_{fr_a} + S^m_{Term\,a} + S^m_{I_a} + S^m_{R_a} + S^m_{CX_a}\right) h_z \sqrt{g} (+S^m_{eirene,a} h_z \sqrt{g})(+S^m_{BC,a} h_z \sqrt{g})$ - the sum of contributions from:

$S^m_{CX_a} h_z \sqrt{g}$ ion-neutral ion-ion charge-exchange friction (***b2stcx_smq000…ns-1***),

$S^m_{I_a} h_z \sqrt{g}$ - ionization (***b2stel_smq_ion***) , $S^m_{R_a} h_z \sqrt{g}$ - recombination (***b2stel_smq_rec000…ns-1***),

$S^m_{fr,ia} h_z \sqrt{g}$ -friction interaction with different other ion species (***b2npmo_smofria000…ns-1***)

$S^m_{fr,ea} h_z \sqrt{g}$ -friction interaction with electrons (***b2npmo_smofrea000…ns-1***)

$S^m_{Term,ia} h_z \sqrt{g}$ -thermal force with ions (***b2npmo _smotfia000…ns-1***)

$S^m_{Term,ea} h_z \sqrt{g}$ -thermal force with electrons (***b2npmo _smotfea000…ns-1***)

$S^m_{eirene,a} h_z \sqrt{g}$ -ionization from, recombination to, and charge-exchange with neutrals in case of EIRENE turned on ***b2stbr_smo_eir001…ns-1***.

$S^m_{BC,a} h_z \sqrt{g}$ - source in the guard cells due to boundary conditions.(Zero in the volume big cells.) ***b2stbc_phys_smo000…ns-1***.

$$S^m_{a\|} = \begin{cases} 0, & a \neq 1, \\ \frac{\partial}{h_z \sqrt{g} \partial x} \left(\frac{h_z \sqrt{g}}{h_x} \frac{4}{3} \eta_{ax} \frac{\partial \ln\left(h_z B^{\frac{1}{2}}\right)}{B^{\frac{1}{2}} h_x \partial x} \right) B^{\frac{1}{2}} V_{\|a} + B^{\frac{3}{2}} b_x \frac{\partial}{h_x \partial x} \left(\alpha^{NEO} \frac{b_x \tau_a}{B^2} \frac{\partial \left(B^{\frac{1}{2}} \tilde{q}^{(0)} \right)}{h_x \partial x} \right) & , \ a = 1 \end{cases}$$

First and second terms of this viscosity momentum source are ***b2npmo_smovv001*** and ***b2npmo_smovh001*** respectively.

$$\tilde{q}^{(0)} = q^{(1)} arctg\left(\frac{q^{(0)}}{q^{(1)}}\right), \quad q^{(1)} = c_{q1} n_i T_i \sqrt{\frac{T_i + T_e}{m_i}}$$

$c_{q1} = 0.5$ or {'b2siav_cqip1' '0.1'} in b2mn.dat

The three-point smoothing procedure is imposed twice on $\tilde{q}^{(0)}$ along *x*-direction for *a*=1

In case {'b2siav_style_qip' '0'} in b2mn.dat

$$q^{(0)} = q^{(0)}_{a\|} + \frac{B}{B_x} q^{(dia)}_{ax} = \begin{bmatrix} 0 & \textit{outside the separatrix} \\ -\frac{5}{2} \frac{B_z B}{B_x < B^2 >} \frac{nT_i}{e} \frac{\partial T_i}{h_y \partial y} & \textit{inside the separatrix} \end{bmatrix}$$

In case {'b2siav_style_qip' '1'} in b2mn.dat

$$q^{(0)} = q_{a\|}^{(0)} + \frac{B}{B_x} q_{ax}^{(dia)} = \begin{cases} 0 & \textit{outside the separatrix} \\ -\tilde{\tilde{k}}_{ix}^{(CL)} b_x \frac{\partial T_i}{h_x \partial x} - \frac{5}{2} \frac{B_z B}{B_x B^2} \frac{nT_i}{e} \frac{\partial T_i}{h_y \partial y} & \textit{inside the separatrix} \end{cases}$$

Plateau and banana correction 'b2trcl_lluciani' '3'

$$\tilde{\tau}_a = \frac{\tau_a}{1 + \dfrac{15{,}12 \cdot 10^{16} \left(\dfrac{T_i}{e}\right)^2}{c(y) \sum_{a=0}^{n_s - 1} z_a^2 n_a}} \cdot \frac{1}{1 + \dfrac{15{,}12 \cdot 10^{16} \cdot \varepsilon^{3/2} \left(\dfrac{T_i}{e}\right)^2}{c(y) \sum_{a=0}^{n_s - 1} z_a^2 n_a}}$$

$$\alpha^{NEO} = \frac{8}{15} \left(\frac{-0.17 + 1.05 \nu_*^{1/2} + 2.7 \nu_*^2 \varepsilon^3}{1 + 0.7 \nu_*^{1/2} + \nu_*^2 \varepsilon^3} - 1 \right)$$ gives the correct coefficient before the temperature gradient in the neoclassical electric field in the plateau and banana regime

$$\nu_* = \frac{c(y)}{2\pi \varepsilon^{3/2} \tau_a \sqrt{T_i / m_a}}, \quad c(y) = \oint h_x b_x^{-1} dx, \quad \varepsilon = \frac{\left(\oint h_x dx\right)^2}{\oint \sqrt{g} / h_y dx}$$

$$\Gamma_{ax}^m = \begin{cases} m_a V_{\|a} \Gamma_{ax} + m_a V_{\|a} \Gamma_{ax}^{(dia)} - \eta_{ax} \dfrac{\partial V_{\|a}}{h_x \partial x}, & a \neq 1 \\ m_a V_{\|a} \Gamma_{ax} + m_a V_{\|a} \Gamma_{ax}^{(dia)} + \dfrac{4}{3} \eta_{ax} \dfrac{\partial \ln h_z}{h_x \partial x} V_{\|a} - \dfrac{4}{3} \eta_{ax} \dfrac{\partial V_{\|a}}{h_x \partial x}, & a = 1 \end{cases}$$

Here the contribution from classic parallel viscosity is included.

$$\left(\Gamma_{ax} + \Gamma_{ax}^{(dia)}\right) \frac{h_z \sqrt{g}}{h_x} = \left(b_x V_{\|a} n_a + V_{ax}^{(E)} n_a + 2 \tilde{V}_{ax}^{(dia)} n_a + V_a^{corr\_dpc} n_a - D_a^n \frac{1}{h_x} \frac{\partial n_a}{\partial x} - \tilde{D}_{ax}^p \frac{1}{h_x} \frac{\partial p_a}{\partial x} \right) \frac{h_z \sqrt{g}}{h_x}$$ -

b2tfnb_fnb_fcorx000…ns-1

$-D_a^n \frac{1}{h_x} \frac{\partial n_a}{\partial x} \frac{h_z \sqrt{g}}{h_x}$ - ***b2tfnb_dPat_2diagradnax001…ns-1*** here diffusion coefficient is modified according to convective flux with $2\tilde{V}_x^{(dia)}$

$\frac{h_z \sqrt{g}}{h_x} \left(\frac{4}{3} \eta_{ax} \frac{\partial \ln h_z}{h_x \partial x} \right)$ - ***b2npmo_flubvx001*** convective flow addition due to parallel viscosity, only for main sort of ions

$-\frac{h_z \sqrt{g}}{h_x} \frac{4}{3} \eta_{ax} \frac{\partial V_{\|a}}{h_x \partial x}$ for $a = 1$ and $-\frac{h_z \sqrt{g}}{h_x} \eta_{ax} \frac{\partial V_{\|a}}{h_x \partial x}$ for $a \neq 1$ - ***b2urmo_etaPat_graduax000…ns-1***

The Patankar correction is applied for η_{ax}

$$\Gamma_{ay}^m = m_a V_{\|a} \Gamma_{ay} + m_a V_{\|a} \Gamma_{ay}^{(dia)} - \eta_{ay} \frac{\partial V_{\|a}}{h_y \partial y},$$

$$\left(\Gamma_{ay}+\Gamma_{ay}^{(dia)}\right)\frac{h_z\sqrt{g}}{h_y}=\begin{cases} V_{ay}^{(E)}n_a+2\tilde{V}_{ay}^{(dia)}n_a+V_{ay}^{(AN)}n_a+\dfrac{j_y^{(AN)}+j_y^{(in)}+\tilde{j}_y^{(vis\|)}}{e}n_a-D_a^n\dfrac{1}{h_y}\dfrac{\partial n_a}{\partial y}-\tilde{D}_{ay}^p\dfrac{1}{h_y}\dfrac{\partial p_a}{\partial y} & a=1\\ V_{ay}^{(E)}n_a+2\tilde{V}_{ay}^{(dia)}n_a+V_{ay}^{(AN)}n_a-D_a^n\dfrac{1}{h_y}\dfrac{\partial n_a}{\partial y}-\tilde{D}_{ay}^p\dfrac{1}{h_y}\dfrac{\partial p_a}{\partial y} & a\neq 1\end{cases}$$

- ***b2tfnb_fnb_fcory000…ns-1***

$-\dfrac{h_z\sqrt{g}}{h_y}\eta_{ay}\dfrac{\partial V_{\|a}}{h_y\partial y}$ - ***b2urmo_etaPat_graduay01*** the Patankar correction is applied for η_{ay}

$\eta_{ax}=\eta_{ANa}+\tilde{\tilde{\eta}}_{CLax}$

$\eta_{ANa}\dfrac{h_z\sqrt{g}}{h_x^2}$ - ***b2trno_cvsahzx000…ns-1*** it coincides for neutrals with their full viscosity (no flux limiting is applied), for ions that is anomalous perpendicular viscosity.

$\eta_{ANa}=n_a m_a \nu_a^{visc}$

ν_a^{visc} - ***b2tqna_vsa000…ns-1*** - at the center of cell

Classical viscosity:

…after flux-limiting and neoclassical corrections

$\tilde{\tilde{\eta}}_{CLax}=c_{\eta_{ax}^{(CL)}}\tilde{\eta}_{CLax}$ $\qquad$ $\tilde{\eta}_{CLax}\dfrac{h_z\sqrt{g}}{h_x^2}$ - ***b2trcl_luciani_fllim_cvsahzx001…ns-1*** for $z_a\neq 0$

$$c_{\eta_{ax}^{(CL)}}=\begin{cases}1, & \textit{only for inner flux surfaces}\\ \dfrac{1}{1+\dfrac{\dfrac{\sqrt{g}}{h_x}\tilde{\eta}_{CLax}\left|\dfrac{\partial V_{\|a}}{h_x\partial x}\right|}{V_{m_a}^{(\max)}}}, & \textit{for all others cases}\end{cases}$$

$$V_{m_a}^{(\max)}=c_{f\lim_2}\frac{\sqrt{g}}{h_x}b_x n_a T_i$$

…after neoclassical correction

$\tilde{\eta}_{CLax}=c_{\eta_{ax}^{(CL)}}^{(Luciani)}\eta_{CLax}$ $\qquad$ $\eta_{CLax}\dfrac{h_z\sqrt{g}}{h_x^2}$ - ***b2trcl_luciani_cvsahzx001…ns-1***

For plateau and banana correction ‘b2trcl_lluciani’ ‘3’

$$c_{\eta_{ax}^{(CL)}}^{(Luciani)}=\begin{cases}\dfrac{1}{1+\dfrac{15{,}12\cdot 10^{16}\left(\dfrac{T_i}{e}\right)^2}{c(y)\sum\limits_{a=0}^{n_s-1}z_a^2n_a}}\cdot\dfrac{1}{1+\dfrac{15{,}12\cdot 10^{16}\cdot\varepsilon^{3/2}\left(\dfrac{T_i}{e}\right)^2}{c(y)\sum\limits_{a=0}^{n_s-1}z_a^2n_a}}, & \textit{only for inner flux surfaces,}\\ 1, & \textit{for all others cases}\end{cases}$$

$$c(y)=\oint h_x b_x^{-1}dx\,,\qquad \varepsilon=\frac{\left(\oint h_x dx\right)^2}{\oint\sqrt{g}\,/\,h_y dx}$$

…initially

$$\eta_{CLax}=\begin{cases}0, & z_a=0\\ \sqrt{2}\eta T_i n_a\tau_a b_x^2, & z_a\neq 0\end{cases},\qquad \eta=0.96\qquad \eta_{CLax}\frac{h_z\sqrt{g}}{h_x^2}$$ - ***b2trcl_cvsahzx001…ns-1***

$$\tau_a=\frac{3}{4\sqrt{2\pi}}\frac{\sqrt{m_a}T_i^{\frac{3}{2}}}{z_a^2 z_{eff}n_e\Lambda}\left(\frac{4\pi\varepsilon_0}{e^2}\right)^2$$ - ***b2tqca_taua001…n$_s$-1***

$$\eta_{ay}=\eta_{ANa}+\eta_{CLay},\quad \eta_{ay}^{(CL)}=0\qquad \eta_{ay}\frac{h_z\sqrt{g}}{h_y^2}$$ - ***b2trno_cvsahzy***

ELECTRON HEAT BALANCE

$$\begin{aligned}&\frac{3}{2}\frac{\partial n_eT_e}{\partial t}+\frac{1}{\sqrt{g}}\frac{\partial}{\partial x}\left(\frac{\sqrt{g}}{h_x}\tilde{q}_{ex}\right)+\frac{1}{\sqrt{g}}\frac{\partial}{\partial y}\left(\frac{\sqrt{g}}{h_y}\tilde{q}_{ey}\right)+\frac{n_eT_e}{\sqrt{g}}\frac{\partial}{\partial x}\left(\frac{\sqrt{g}b_x}{h_x}V_{e\parallel}\right)=\\&=-Q_\Delta+Q_R+Q_{eirene,e}+Q_{BC}^{(e)}+n_eT_eB\frac{1}{h_xh_y}\left(\frac{\partial\phi}{\partial y}\frac{\partial}{\partial x}\left(\frac{1}{B^2}\right)-\frac{\partial\phi}{\partial x}\frac{\partial}{\partial y}\left(\frac{1}{B^2}\right)\right)+j_x\hat{E}_x,\end{aligned}\tag{3}$$

T_e - ***b2nph9_te***

The residuals of the iteration of this equation are ***b2nph9_reshe***

$\frac{\sqrt{g}}{h_{x,y}}\tilde{q}_{ex,y}$ - ***b2nph9_fhex,y***

$$\left(Q_R+Q_{eirene,e}+Q_{BC}^{(e)}+n_eT_eB\frac{1}{h_xh_y}\left(\frac{\partial\phi}{\partial y}\frac{\partial}{\partial x}\left(\frac{1}{B^2}\right)-\frac{\partial\phi}{\partial x}\frac{\partial}{\partial y}\left(\frac{1}{B^2}\right)\right)+Q_e^{fr}-\frac{n_eT_e}{\sqrt{g}}\frac{\partial}{\partial x}\left(\frac{\sqrt{g}b_x}{h_x}V_{e\parallel}\right)\right)\sqrt{g}$$ - ***b2nph9_she***

$Q_R\sqrt{g}$ - ***b2stel_she_rad-*** includes both ionization and radiation losses, bremsstrahlung radiation due to the electron collisions with ions, source due to collisions with neutrals if EIRENE is turned off

b2stel_rqbrm001…- bremsstrahlung radiation due to the electron collisions with ions of chosen sort

b2stel_rqrad 001…- line radiation due to the electron collisions with ions of chosen sort

$$Q_\Delta=c_{eqp}\Lambda e^{\frac{3}{2}}\frac{n_i}{\sum\limits_{a=0}^{n_s-1}\frac{z_a^2m_p}{m_a}n_a}\frac{n_en_i}{T_e^{\frac{3}{2}}}(T_e-T_i),\quad c_{eqp}=4{,}8\cdot 10^{-15},\quad \Lambda=12\,,\qquad Q_\Delta\sqrt{g}$$ - ***b2nph9_shei*** thermal exchange between ions and electrons

$Q_{eirene,e}\sqrt{g}$ -***b2stbr_she_eir*** electron heat source from EIRENE containing neutral ionization losses and neutral excitation losses

$Q_e^{fr}\sqrt{g} = \sqrt{g}V_{e\parallel}\sum_{\substack{a=0\\ z_a\neq 0}}^{n_s-1}\left(S^m_{fr_{ea}} + S^m_{Term\,ea}\right)$ - ***b2sihs__shefr*** contains Ohmic heating; in old version $Q_e^{fr} = j_x\hat{E}_x$

$-\frac{n_eT_e}{\sqrt{g}}\frac{\partial}{\partial x}\left(\frac{\sqrt{g}b_x}{h_x}V_{e\parallel}\right)\sqrt{g}$ - ***b2sihs__shedu***

$n_eT_eB\frac{1}{h_xh_y}\left(\frac{\partial\phi}{\partial y}\frac{\partial}{\partial x}\left(\frac{1}{B^2}\right) - \frac{\partial\phi}{\partial x}\frac{\partial}{\partial y}\left(\frac{1}{B^2}\right)\right)\sqrt{g}$ - ***b2sihs__shedd*** divergence of E×B drift multiplied to electron pressure

$Q_{BC}^{(e)}\sqrt{g}$ source in the guard cells due to boundary conditions.(Zero in the volume big cells.) ***b2stbc_phys_she***

$-\frac{3}{2}\frac{\partial n_eT_e}{\partial t}\sqrt{g}$ - ***b2srdt_shedt***

For numerical stabilization ***b2srst_shest*** appears. Normally it is negligible.

$$\tilde{q}_{ex} = \frac{3}{2}\Gamma_{ex}T_e - \kappa_{ex}\frac{1}{h_x}\frac{\partial T_e}{\partial x} + \alpha_{ex}T_e\hat{E}_x - q_{ex}^{P.Sch}$$

$\frac{\sqrt{g}}{h_{x,y}}\frac{3}{2}\Gamma_{ex,y}T_e$ - ***b2tfhe__qe_32x,y***

$q_{ex}^{P.Sch} = -\frac{5}{2}D_{exy}\frac{1}{h_y}\frac{\partial n_eT_e^2}{\partial y}$, $\frac{\sqrt{g}}{h_{x,y}}q_{ex,y}^{P.Sch}$ - ***b2tfhe_fhePSchx,y*** effective heat flow giving the same divergence as grad B heat flow

$-\frac{\sqrt{g}}{h_{x,y}}\kappa_{ex,y}\frac{1}{h_{x,y}}\frac{\partial T_e}{\partial x, y}$ - ***b2tfhe__qe_ke_gTx,y*** heat conductivity modified by Patankar correction

$\alpha_{ex,y}T_e\hat{E}_{x,y}$ - ***b2tfhe__qe_alphaTehx,y***

$\Gamma_{ex} = \sum_{a=0}^{n_s-1}z_a\Gamma_{ax}^{(he)} - \frac{j_x^{(\parallel)}}{e}$,

$\Gamma_{ax}^{(he)} = (b_xV_{a\parallel} + V_{ax}^{(E)} + V_a^{corr\_dpc})n_a - \frac{5}{3}\left[D_a^n\frac{1}{h_x}\frac{\partial n_a}{\partial x} + \tilde{D}_{ax}^p\frac{1}{h_x}\frac{\partial p_a}{\partial x}\right]$, $\Gamma_{ax}^{(he)}\frac{\sqrt{g}}{h_x}$ - ***b2tfnb_fnb_hex***

Here D_a^n is modified by Patankar correction according to convective flows in $\Gamma_{ax}^{(he)}$

$\kappa_{ex} = \kappa_e^{(AN)} + \tilde{\kappa}_{ex}^{(CL)}$, $\kappa_{ex,y}\frac{\sqrt{g}}{h_{x,y}^2}$ -modified by Patankar correction ***b2tfhe__chcex,y_Pat,***

$\kappa_e^{(AN)}\frac{\sqrt{g}}{h_{x,y}^2}$ - ***b2trno_chcex,y*** anomalous electron thermal conductivity coefficient before the correction

$$\kappa_e^{(AN)} = n_e \chi_e^{(AN)}$$

$\chi_e^{(AN)}$ - ***b2tqna_hce0*** at the centers of cells

Classical thermal conductivity of electrons

…after the flux limiting and neoclassical correction

$$\tilde{\tilde{\kappa}}_{ex}^{(CL)} = c_{k_{ex}}^{(F\lim)} \tilde{\kappa}_{ex}^{(CL)} \qquad \tilde{\tilde{\kappa}}_{ex}^{(CL)} \frac{\sqrt{g}}{h_x^{\,2}}$$ - ***b2trcl_luciani_fllim_chcex***

$$c_{k_{ex}}^{(F\lim)} = \begin{cases} 1, & \textit{only for inner flux surfaces} \\ \dfrac{1}{1 + \dfrac{\dfrac{\sqrt{g}}{h_x} \tilde{\kappa}_{ex}^{(CL)} \left| \dfrac{\partial T_e}{h_x \partial x} \right|}{V_{h_e}^{(\max)} n_e T_e}}, & \textit{for other flux surfaces} \end{cases}$$

$$V_{h_e}^{(\max)} = c_{f\lim_0} \frac{\sqrt{g}}{h_x} b_x \sqrt{\frac{T_e}{m_e}},$$

After neoclassical correction

$$\tilde{\kappa}_{ex}^{(CL)} = c_{\kappa_{ex}^{(CL)}}^{(Luciani)} \kappa_{ex}^{(CL)} b_x^2, \qquad c_{\kappa_{ex}^{(CL)}}^{(Luciani)} = 1 \qquad \tilde{\kappa}_{ex}^{(CL)} \frac{\sqrt{g}}{h_x^{\,2}}$$ - ***b2trcl_luciani_chcex***

For plateau and banana correction 'b2trcl_lluciani' '3'

$$c_{\kappa_{ex}^{(CL)}}^{(Luciani)} = \begin{cases} 0.58\varepsilon^{3/2} / K_1, & \text{only for inner flux surfaces} \\ 1, & \text{for other flux surfaces} \end{cases}$$

$$K_1 = \frac{1}{1 + 2.01 \cdot \nu_{*e}^{1/2} + 1.53 \cdot \nu_{*e}} + \frac{0.52 \cdot \varepsilon^3 \cdot \nu_{*e}}{1 + 0.89 \cdot \varepsilon^{3/2} \cdot \nu_{*e}}$$

$$\nu_{*e} = \frac{c(y)}{2\pi\varepsilon^{3/2} \tau_e \sqrt{T_e / m_e}},$$

…initially

$$\kappa_{ex}^{(CL)} = \frac{5}{2} \frac{T_e}{m_e} f_{k_e}(z_{eff}) n_e \tau_e, \qquad \kappa_{ex}^{(CL)} \frac{\sqrt{g}}{h_x^{\,2}} b_x^2$$ - ***b2trcl_chcex***

$$z_{eff} = \frac{\sum_{a=0}^{n_s-1} z_a^2 n_a}{n_e}$$ ***zeff***,

$$\tau_e = \frac{3}{4\sqrt{2\pi}} \frac{\sqrt{m_e} T_e^{\frac{3}{2}}}{z_{eff} n_e \Lambda} \left(\frac{4\pi\varepsilon_0}{e^2} \right)^2,$$ - ***b2tqce_taue***

Λ - Coulomb logarithm,

$$V_{e\parallel} = \frac{1}{e n_e} \left[\sum_{a=0}^{n_s-1} z_a e n_a V_{a\parallel} - \frac{j_x^{(\parallel)}}{b_x} \right]$$ - ***b2sifrtf_ue***,

$$j_x^{(\|)} = \sigma_\| \left(\frac{1}{en_e h_x}\frac{\partial n_e T_e}{\partial x} - \frac{1}{h_x}\frac{\partial \varphi}{\partial x} \right) - \alpha_{ex}\frac{1}{h_x}\frac{\partial T_e}{\partial x} - \frac{eb_x}{z_{eff}}\sum_{a=0}^{n_s-1} n_a V_{a\|}\left(z_a^2 - z_a z_{eff}\right)$$

$$j_x^{(\|)} = \sigma_\| \hat{E}_x - \alpha_{ex}\frac{1}{h_x}\frac{\partial T_e}{\partial x},$$

$$\hat{E}_x = \frac{1}{en_e h_x}\frac{\partial n_e T_e}{\partial x} - \frac{1}{h_x}\frac{\partial \phi}{\partial x}$$

$$\sigma_\| = c_{\sigma_{CL\|}}^{(Luciani)}\sigma_{CL\|}$$ $\sigma_\| \frac{\sqrt{g}}{h_x^{\ 2}}$ - ***b2trcl_luciani_csigx***

For plateau and banana correction 'b2trcl_lluciani' '3'

$$c_{\sigma_{CL\|}}^{(Luciani)} = \begin{cases} 0.58\varepsilon^{3/2}/K_1, & \text{only for inner flux surfaces} \\ 1, & \text{for other flux surfaces} \end{cases}$$

$$\sigma_{CL\|} = b_x^2 \frac{e^2}{m_e} f_{\sigma_e}\left(z_{eff}\right)\tau_e n_e z_{eff}$$ $\sigma_{CL\|}\frac{\sqrt{g}}{h_x^{\ 2}}$ - ***b2trcl_csigx***

$$\alpha_{ex} = c_{\alpha_{ex}}^{(F\lim)}\tilde{\alpha}_{CLx},$$ $\alpha_{ex}\frac{\sqrt{g}}{h_x^{\ 2}}$ - ***b2trcl_luciani_fllim_calfx***

$$c_{\alpha_{ex}}^{(F\lim)} = \begin{cases} 1, & \text{only for inner flux surfaces} \\ \dfrac{1}{1 + \left| \dfrac{\alpha_{CLx}\frac{1}{h_x}\frac{\partial T_e}{\partial x}}{c_{\text{flime}}\, j_{e\max}} \right|}, & \text{for other flux surface} \end{cases}$$ $$j_{e\max} = |b_x| en_e\sqrt{\frac{T_e}{m_e}}$$

$$\tilde{\alpha}_{CLx} = c_{\alpha_{ex}^{(CL)}}^{(Luciani)}\alpha_{CLx},$$ $\tilde{\alpha}_{CLx}\frac{\sqrt{g}}{h_x^{\ 2}}$ - ***b2trcl_luciani_calfx***

$$\alpha_{CLx} = b_x^2\frac{e}{m_e} f_\alpha\left(z_{eff}\right)\tau_e n_e z_{eff},$$ $\alpha_{CLx}\frac{\sqrt{g}}{h_x^{\ 2}}$ - ***b2trcl_calfx***

$$\tilde{q}_{ey} = \frac{3}{2}\Gamma_{ey}T_e - \kappa_e^{(AN)}\frac{1}{h_y}\frac{\partial T_e}{\partial y} - q_{ey}^{P.Sch}$$

$$\Gamma_{ey} = \sum_{a=0}^{n_s-1} z_a \Gamma_{ay}^{(he)}$$

$$q_{ey}^{P.Sch} = \frac{5}{2}D_{eyx}\frac{1}{h_x}\frac{\partial n_e T_e^2}{\partial x}$$

$$\Gamma_{ay}^{(he)} = (\frac{5}{3}V_{ay}^{(AN)} + V_{ay}^{(E)})n_a - \frac{5}{3}\left[D_a^n\frac{1}{h_y}\frac{\partial n_a}{\partial y} + \tilde{D}_{ay}^p\frac{1}{h_y}\frac{\partial p_a}{\partial y}\right]$$ $\Gamma_{ay}^{(he)}\frac{\sqrt{g}}{h_y}$ - ***b2tfnb_fnb_hey***

ION HEAT BALANCE

$$\frac{3}{2}\frac{\partial n_i T_i}{\partial t}+\frac{1}{\sqrt{g}}\frac{\partial}{\partial x}\left(\frac{\sqrt{g}}{h_x}\tilde{q}_{ix}\right)+\frac{1}{\sqrt{g}}\frac{\partial}{\partial y}\left(\frac{\sqrt{g}}{h_y}\tilde{q}_{iy}\right)+\sum_{a=0}^{n_s-1}\frac{n_a T_i}{\sqrt{g}}\frac{\partial}{\partial x}\left(\frac{\sqrt{g}}{h_x}V_{a\parallel}b_x\right)=$$

$$=Q_\Delta+Q_{F_{ab}}+Q_I^{(i)}+Q_R^{(i)}+Q_{eirene,i}+Q_{BC}^{(i)}+c_{E\times B}T_i B\frac{1}{h_x h_y}\left(\frac{\partial\phi}{\partial y}\frac{\partial}{\partial x}\left(\frac{1}{B^2}\right)-\frac{\partial\phi}{\partial x}\frac{\partial}{\partial y}\left(\frac{1}{B^2}\right)\right)\sum_{\substack{a=0\\ z_a\neq 0}}^{n_s-1}n_a+ \qquad (4)$$

$$+\sum_{\substack{a=0\\ z_a\neq 0}}^{n_s-1}\left(\frac{4}{3}\eta_{ax}\left(\frac{\partial V_{a\parallel}}{h_x\partial x}\right)^2+\eta_{ANa}\left(\frac{\partial V_{a\parallel}}{h_y\partial y}\right)^2\right),$$

T_i - ***b2nph9_ti***

The residuals of the iteration of this equation are ***b2nph9_reshi***

$\frac{\sqrt{g}}{h_{x,y}}\tilde{q}_{ix,y}$ - ***b2nph9_fhix,y***

$-\sqrt{g}\,\frac{3}{2}\frac{\partial n_i T_i}{\partial t}$ - ***b2srdt_shidt***

$$(Q_{F_{ab}}+Q_I^{(i)}+Q_R^{(i)}+Q_{CX}^{(i)}+Q_{eirene,i}+Q_{BC}^{(i)}+c_{E\times B}T_i B\frac{1}{h_x h_y}\left(\frac{\partial\phi}{\partial y}\frac{\partial}{\partial x}\left(\frac{1}{B^2}\right)-\frac{\partial\phi}{\partial x}\frac{\partial}{\partial y}\left(\frac{1}{B^2}\right)\right)\sum_{\substack{a=0\\ z_a\neq 0}}^{n_s-1}n_a+ \quad -$$

$$+\sum_{\substack{a=0\\ z_a\neq 0}}^{n_s-1}\left(\frac{4}{3}\eta_{ax}\left(\frac{\partial V_{a\parallel}}{h_x\partial x}\right)^2+\eta_{ANa}\left(\frac{\partial V_{a\parallel}}{h_y\partial y}\right)^2\right)-\sum_{a=0}^{n_s-1}\frac{n_a T_i}{\sqrt{g}}\frac{\partial}{\partial x}\left(\frac{\sqrt{g}}{h_x}V_{a\parallel}b_x\right))\sqrt{g},$$

b2nph9_shi

$\sqrt{g}\,c_{E\times B}T_i B\frac{1}{h_x h_y}\left(\frac{\partial\phi}{\partial y}\frac{\partial}{\partial x}\left(\frac{1}{B^2}\right)-\frac{\partial\phi}{\partial x}\frac{\partial}{\partial y}\left(\frac{1}{B^2}\right)\right)\sum_{\substack{a=0\\ z_a\neq 0}}^{n_s-1}n_a$ -***b2sihs__shidd*** divergence of E×B drift multiplied to ion pressure

$-\sqrt{g}\sum_{a=0}^{n_s-1}n_a T_i\frac{\partial}{\partial x}\left(\frac{\sqrt{g}}{h_x}V_{a\parallel}b_x\right)$ **- *b2sihs__shidu***

$\sqrt{g}\sum_{\substack{a=0\\ z_a\neq 0}}^{n_s-1}\frac{4}{3}\eta_{ax}\left(\frac{\partial V_{a\parallel}}{h_x\partial x}\right)^2$ - ***b2sihs__shivc*** viscous heating due to classical viscosity

$\sqrt{g}\sum_{\substack{a=0\\ z_a\neq 0}}^{n_s-1}\eta_{ANa}\left(\frac{\partial V_{a\parallel}}{h_y\partial y}\right)^2$ - ***b2sihs__shiva*** viscous heating due to anomalous viscosity

$Q_{F_{ab}}\sqrt{g} = -\sqrt{g}\sum_{\substack{a=0\\ z_a\neq 0}}^{n_s-1} V_{a\parallel}\left(S^m_{fr_a} + S^m_{Term\,a}\right)$- heating because of friction between ion species and friction with electrons–***b2sihs__shifr***

$Q_I^{(i)}\sqrt{g}$ - heat source due to ionization–***b2stel_shi_ion***

$Q_R^{(i)}\sqrt{g}$ - heat source due to recombination–***b2stel_shi_rec***

$Q_{CX}^{(i)}\sqrt{g}$ - heat source due to charge-exchange - ***b2stcx_shi***

$Q_{eirene,i}\sqrt{g}$ -***b2stbr_shi_eir*** ion heat source from EIRENE, including ionization heat source

$Q_{BC}^{(i)}\sqrt{g}$ **-** source in the guard cells due to boundary conditions.(Zero in the volume big cells.) ***b2stbc_phys_shi***

For numerical stabilization ***b2srst_shist*** appears. Normally it is negligible.

$\tilde{q}_{ix} = \frac{3}{2}\Gamma_{ix}T_i - \kappa_{ix}\frac{1}{h_x}\frac{\partial T_i}{\partial x} - q_{ix}^{P.Sch}$, effective poloidal ion heat flow

$\frac{\sqrt{g}}{h_{x,y}}\frac{3}{2}\Gamma_{ix,y}T_i$ - ***b2tfhi__qi_32x,y***

$q_{ix}^{P.Sch} = \frac{5}{2}\sum_{\substack{a=0\\ z_a\neq 0}}^{n_s-1} D_{axy}\frac{1}{h_y}\frac{\partial n_a T_i^2}{\partial y}$ $\frac{\sqrt{g}}{h_{x,y}}q_{ix,y}^{P.Sch}$ - ***b2tfhi__fhiPSchx,y*** effective heat flow giving the same divergence as grad B heat flow

$-\kappa_{ix,y}\frac{1}{h_{x,y}}\frac{\partial T_i}{\partial x,y}$ - ***b2tfhi__qi_ki_gTx,y*** conductive flow, Patankar correction is applied

$\Gamma_{ix} = \sum_{a=0}^{n_s-1}\Gamma_{ax}^{(he)}$

$\kappa_{ix} = \kappa_{ix}^{(AN)} + \tilde{\tilde{\kappa}}_{ix}^{(CL)} b_x^2$,

$\kappa_{ix,y}\frac{\sqrt{g}}{h_{x,y}^{\;2}}$ -modified by Patankar correction ***b2tfhi__chcix,y_Pat,***

$k_{ix}^{(AN)} = \sum_{a=0}^{n_s-1}\tilde{k}_{ax}^{(AN)} = \sum_{a=0}^{n_s-1}\chi_a^{(AN)} n_a$, ion anomalous thermal conductivity coefficient for ions or flux-limited thermal conductivity for neutrals

$\chi_a^{(AN)}$- ***b2tqna_hcib000.. ns-1***

$$\tilde{k}_{ax}^{(AN)} = \begin{cases} k_{ax}^{(AN)}, & z_a \neq 0 \\ \dfrac{k_{ax}^{(AN)}}{\left(1+\left(\dfrac{k_a^{(AN)}\left|\dfrac{\partial T_i}{h_x \partial x}\right|}{\alpha V_{h_i} n_a T_i}\right)^{\gamma_2}\right)^{\frac{1}{\gamma_2}}}, & z_a = 0 \end{cases} \qquad V_{h_i} = \sqrt{\frac{2T_i}{m_a}}$$

$\kappa_{ix,y}^{(AN)} \dfrac{\sqrt{g}}{h_{x,y}^{\ 2}}$ - ***b2trno_chcix,y***

Classical ion thermal conductivity

…after flux-limiting and neoclassical correction

$$\tilde{\tilde{k}}_{ix}^{(CL)} = c_{k_{ix}}^{(F\lim)} \tilde{k}_{ix}^{(CL)} \qquad \tilde{\tilde{\kappa}}_{ix}^{(CL)} \frac{\sqrt{g}}{h_x^{\ 2}}$$ - ***b2trcl_luciani_fllim_chcix***

$$c_{k_{ix}}^{(F\lim)} = \begin{cases} 1, & \textit{for inner flux surfaces} \\ \dfrac{1}{1+\dfrac{\dfrac{\sqrt{g}}{h_x} k_{ix}^{(CL)} \left|\dfrac{\partial T_i}{h_x \partial x}\right|}{V_{h_i}^{(\max)} n_i T_i}}, & \textit{for other flux surfaces} \end{cases},$$

$$V_{h_i}^{(\max)} = c_{f\lim_1} \frac{\sqrt{g}}{h_x} |b_x| \sqrt{\frac{T_i}{m_p}} \frac{\sum_{a=0}^{n_s-1} n_a \sqrt{\dfrac{m_p}{m_a}}}{n_i},$$

$c_{f\lim_1} = 0{,}15$ (or other value, prescribed in b2mn.dat)

…after neoclassical correction

$$\tilde{\kappa}_{ix}^{(CL)} = c_{k_{ix}}^{(Luciani)} \kappa_{ix}^{(CL)}, \qquad \tilde{\kappa}_{ix}^{(CL)} \frac{\sqrt{g}}{h_x^{\ 2}}$$ - ***b2trcl_luciani_chcix***

For plateau and banana correction 'b2trcl_lluciani' '3'

$c_{k_{ix}}^{(Luciani)} = 1.6 \cdot \varepsilon^{3/2} / K_2$,

$$K_2 = \frac{0.66 + 1.88 \cdot \sqrt{\varepsilon} - 1.54 \cdot \varepsilon}{1 + 1.03 \cdot \nu_*^{1/2} + 0.31 \cdot \nu_*} + \frac{1.17 \cdot \varepsilon^3 \cdot \nu_*}{1 + 0.74 \cdot \varepsilon^{3/2} \cdot \nu_*}$$

$$\kappa_{ix}^{(CL)} = b_x^2 \frac{5}{2} \sum_{\substack{a=0 \\ z_a \neq 0}}^{n_s-1} \frac{T_i}{m_a} n_a f_{k_i} \tau_a, \quad f_{k_i} = 2.253, \qquad \kappa_{ix}^{(CL)} \frac{\sqrt{g}}{h_x^{\ 2}}$$ - ***b2trcl_chcix***

For integral modeling ('b2trcl_integral_modeling' '1') ***in the region of the overlap with ASTRA:***

$$\tilde{\kappa}_{ix}^{(CL)} = \frac{25}{4} \left(n_1 T_i \frac{B_z}{B_x h_y} \right)^2 \frac{h_{y(outer\ midplane)}}{e} \left[\oint \frac{\sqrt{g}}{B^2} \left(1 - \frac{B^2}{\langle B^2 \rangle} \right)^2 dx \right] \times K_{ASTRA},$$

Here $\left\langle B^2 \right\rangle = \dfrac{\oint B^2 \sqrt{g}\, dx}{\oint \sqrt{g}\, dx}$

From ASTRA we get T_i / e and $\oint q_i^{NEO} dS$ as a functions of $R_{(outer\, midplane)}$ and calculate

$K_{ASTRA} = \dfrac{-1}{\oint q_i^{NEO} dS} \dfrac{d(T_i / e)}{dR_{(outer\, midplane)}}$ as a function of the flux surface in B2

$\tilde{q}_{iy} = \dfrac{3}{2} \Gamma_{iy} T_i - \kappa_{iy} \dfrac{1}{h_y} \dfrac{\partial T_i}{\partial y} - q_{iy}^{P.Sch} + \dfrac{3}{2} T_i \dfrac{j_y^{(in)} + j_y^{(AN)} + j_y^{(vis\|)}}{e}$ effective radial ion heat flow

$$q_{iy}^{P.Sch} = -\frac{5}{2} \sum_{\substack{a=0 \\ z_a \neq 0}}^{n_s - 1} D_{ayx} \frac{1}{h_x} \frac{\partial n_a T_i^2}{\partial x}$$

$$\Gamma_{iy} = \sum_{a=0}^{n_s - 1} \Gamma_{ay}^{(he)}$$

$$\kappa_{iy} = \kappa_{iy}^{(AN)},$$

$$k_{iy}^{(AN)} = \sum_{a=0}^{n_s - 1} \tilde{k}_{ay}^{(AN)} = \sum_{a=0}^{n_s - 1} \chi_a^{(AN)} n_a,$$

$$\tilde{k}_{ay}^{(AN)} = \begin{cases} k_{ay}^{(AN)}, & z_a \neq 0 \\ \dfrac{k_{ay}^{(AN)}}{\left(1 + \left(\dfrac{\dfrac{\sqrt{g}}{h_y} k_a^{(AN)} \left| \dfrac{\partial T_i}{h_y \partial y} \right|}{\alpha V_{h_i} n_a T_i}\right)^{\gamma_2}\right)^{\frac{1}{\gamma_2}}}, & z_a = 0 \end{cases}$$

CURRENT BALANCE

$$\frac{1}{\sqrt{g}} \frac{\partial}{\partial x} \left(\frac{\sqrt{g}}{h_x} j_x \right) + \frac{1}{\sqrt{g}} \frac{\partial}{\partial y} \left(\frac{\sqrt{g}}{h_y} j_y \right) = 0$$

φ - ***b2npp7_po***

The residuals of the iteration of this equation are ***b2npp7_respo***

$\dfrac{\sqrt{g}}{h_{x,y}} j_{x,y}$ - ***b2nph9_fhix,y*** currents

$$j_x = j_x^{(AN)} + \tilde{j}_x^{(dia)} + j_x^{(in)} + \tilde{j}_x^{(vis\|)} + \tilde{j}_x^{(s)} + j_x^{(\|)},$$

$\dfrac{\sqrt{g}}{h_x} j_x^{(\|)}$ - ***b2tfhe_fch_px***

∇B current:

$$\tilde{j}_x^{(dia)} = B_z \frac{\sum\limits_{\substack{a=0 \\ z_a \neq 0}}^{n_s-1} n_a \left(z_a T_e + T_i\right)}{h_y} \frac{\partial}{\partial y}\left(\frac{1}{B^2}\right), \qquad \frac{\sqrt{g}}{h_{x,y}} \tilde{j}^{(dia)}{}_{x,y}$$ - ***b2tfhe_fchdiax,y***

Inertial current:

$$j_x^{(in)} = \frac{B_z}{2} \frac{\partial}{h_y \partial y}\left(\frac{1}{B^2}\right) \sum_{\substack{a=0 \\ z_a \neq 0}}^{n_s-1} m_a n_a V_{a\|}^2 , \qquad \frac{\sqrt{g}}{h_{x,y}} j^{(in)}{}_{x,y}$$ - ***b2tfhe_fchinertx,y***

Parallel viscosity effective current

$$\tilde{j}_x^{(vis\|)} = -\frac{b_z B_x}{3B^{\frac{1}{2}}} \frac{\partial}{h_y \partial y}\left(\frac{1}{B^2}\right) \sum_{\substack{a=0 \\ z_a \neq 0}}^{n_s-1} \eta_{ax} \frac{\partial\left(\sqrt{B} V_{a\|}\right)}{h_x \partial x}, \qquad \frac{\sqrt{g}}{h_{x,y}} \tilde{j}^{(vis\|)}{}_{x,y}$$ - ***b2tfhe_fchvisparx,y***

Current due to ion-neutral collisions

$$\tilde{j}_x^{(s)} = -\sigma_{a0} b_z^2 \frac{\partial \phi}{h_x \partial x} - \sigma_{a0} b_z^{\,2} \frac{1}{Z_a e n_a} \frac{\partial n_a T_i}{h_x \partial x} + \sigma_{a0} B_z V_{y0}, \; a = 1, \qquad \frac{\sqrt{g}}{h_{x,y}} \tilde{j}^{(s)}{}_{x,y}$$ - ***b2tfhe_fchinx,y***

$$\sigma_{a0} = \frac{m_a n_a \langle V_{a0} \sigma_{ex} \rangle n_0}{2B^2}, \quad a = 1, \qquad \sigma_{a0} b_z^2 \frac{\sqrt{g}}{h_{x,y}^2}$$ - ***b2tral_csiginx01_00***

$$\langle V_{a0} \sigma_{ex} \rangle = 3.2 \cdot 10^{-15} \sqrt{\frac{T_i}{0.026e}}$$

$$V_{y0} = \frac{\Gamma_{0y}}{n_0}$$

$$j_y = j_y^{(AN)} + \tilde{j}_y^{(dia)} + j_y^{(in)} + \tilde{j}_y^{(vis\|)} + j_y^{(vis\perp)} + \tilde{j}_y^{(visq)} + \tilde{j}_y^{(s)} + j_y^{(ST)},$$

Anomalous current, necessary for convergence

$$j_y^{(AN)} = -\sigma_{ANy} \frac{\partial \phi}{h_y \partial y}, \qquad \frac{\sqrt{g}}{h_{x,y}} j_{x,y}^{(AN)}$$ - ***b2tfhe_csig_any***

σ_{ANy} - ***b2tqna_sig0***

$$\tilde{j}_y^{(dia)} = -\frac{B_z \sum\limits_{\substack{a=0 \\ z_a \neq 0}}^{n_s-1} n_a \left(z_a T_e + T_i\right)}{h_x} \frac{\partial}{\partial x}\left(\frac{1}{B^2}\right),$$

$$j_y^{(in)} = -\frac{B_z}{2} \frac{\partial}{h_x \partial x}\left(\frac{1}{B^2}\right) \sum_{\substack{a=0 \\ z_a \neq 0}}^{n_s-1} m_a n_a V_{a\|}^2 ,$$

$$\tilde{j}_y^{(vis\|)} = b_z \frac{B_x}{3B^{\frac{1}{2}}} \frac{\partial}{h_x \partial x}\left(\frac{1}{B^2}\right) \sum_{\substack{a=0 \\ z_a \neq 0}}^{n_s-1} \eta_{ax} \frac{\partial\left(B^{\frac{1}{2}} V_{a\|}\right)}{h_x \partial x},$$

Current associated with perpendicular viscosity

$$j_y^{(vis\perp)} = -\frac{1}{B\sqrt{g}} \frac{\partial}{\partial y}\left(\frac{\sqrt{g}}{h_y} \frac{\eta_{ANa}}{4} \frac{\partial V_{a\perp}}{h_y \partial y}\right), \; a = 1, \qquad \frac{\sqrt{g}}{h_y} j_y^{(vis\perp)}$$ - ***b2tfhe_fchvispery***

$$V_{a\perp} = -\frac{1}{Bh_y}\frac{\partial\phi}{\partial y} - \frac{1}{z_a eBh_y n_a}\frac{\partial n_a T_i}{\partial y}$$

Current due to viscosity associated with heat fluxes

$$\tilde{j}_y^{(visq)} = b_z \frac{B_x}{3B^{\frac{1}{2}}}\frac{\partial}{h_x\partial x}\left(\frac{1}{B^2}\right)\alpha^{NEO}\tau_a \frac{\partial\left(B^{\frac{1}{2}}\tilde{q}^{(0)}\right)}{h_x\partial x},\ a=1$$ - ***b2tfhe_fchvisqy***

$$\tilde{j}_y^{(s)} = -\sigma_{a0}\frac{\partial\phi}{h_y\partial y} - \sigma_{a0}\frac{1}{Z_a en_a}\frac{\partial n_a T_i}{h_y\partial y} - \sigma_{a0}B\,V_{\perp 0},\ \ a=1, \qquad V_{\perp 0} = \frac{\Gamma_{0x}}{n_0}$$

$$\sigma_{ANy} = \left(c_{\sigma_a,0} + c_{\sigma_a,1}\frac{10^{20}}{n_e} + c_{\sigma_a,3}\frac{6{,}2\cdot 10^{-2}T_e}{e|B|} + c_{\sigma_a,4}e^{-c_{\sigma_a,5}\left(\frac{y}{L(x)}\right)^2}\right)e\tilde{n}_e,\ \tilde{n}_e - \textit{density at inner flux surface}$$

$$L(x) = \int_0^{y_{out}} h_y(x,y)dy \qquad c_{\sigma_a,0} = 10^{-6} - 10^{-3},\ c_{\sigma_a,1} = 0,\ c_{\sigma_a,3} = 0$$

Stochastic conductivity current***:***

$$\tilde{j}_y^{(st)} = \begin{cases} -\sigma^{(ST)}\left(\frac{1}{h_y}\frac{\partial\phi}{\partial y} - \frac{T_e}{en_e}\frac{1}{h_y}\frac{\partial n_e}{\partial y} - \frac{0.5}{e}\frac{1}{h_y}\frac{\partial T_e}{\partial y}\right), & y\in\left[-\Delta^{(ST)},0\right], \\ 0,\ y\notin\left[-\Delta^{(ST)},0\right] \end{cases} \qquad \frac{\sqrt{g}}{h_y}\tilde{j}_y^{(st)}$$ - ***b2tfhe_fchstochy***

$$\sigma^{(ST)} = \begin{cases} c_{\sigma^{(ST)}}en_e, & y\in\left[-\Delta^{(ST)},0\right] \\ 0, & y\notin\left[-\Delta^{(ST)},0\right] \end{cases}, \qquad c_{\sigma^{(ST)}} > c_{\sigma_{ANy}}$$

EIRENE output

In this output numeration is different: the sorts of neutrals are numerated separately from ions, so that number 1 corresponds to main particles, 2 – to the first sort of impurities and so on.

b2stbr_dab2_eir001.. – atom density for 01.. sort of atoms
b2stbr_dmb2_eir001.. – atom density for 01.. sort of molecules
b2stbr_pfluxa_eir001..– poloidal flux (particles to meter^2 to second, at the centre of the cell) for 01.. sort of atoms
b2stbr_rfluxa_eir001..– radial flux (particles to meter^2 to second) for 01.. sort of atoms
b2stbr_tfluxa_eir001..– toroidal flux (particles to meter^2 to second) for 01.. sort of atoms
b2stbr_pfluxm_eir001..– poloidal flux (particles to meter^2 to second) for 01.. sort of molecules
b2stbr_rfluxm_eir001..– radial flux (particles to meter^2 to second) for 01.. sort of molecules
b2stbr_tfluxm_eir001..– toroidal flux (particles to meter^2 to second) for 01.. sort of molecules
b2stbr_tab2_eir001..– temperature for 01.. sort of molecules
b2stbr_tmb2_eir001..– temperature for 01.. sort of molecules
b2stbr_fatmx_eir001.. - poloidal flux (particles to meter^2 to second) for 01.. sort of atoms at the face of the cell, interpolation of ***b2stbr_pfluxa_eir001***
b2stbr_fatmy_eir001.. - radial flux (particles to meter^2 to second) for 01.. sort of atoms at the face of the cell, interpolation of ***b2stbr_rfluxa_eir001***
b2stbr_fmolx_eir001.. - poloidal flux (particles to meter^2 to second) for 01.. sort of molecules at the face of the cell, interpolation of ***b2stbr_pfluxm_eir001***

b2stbr_fmoly_eir001.. - radial flux (particles to meter^2 to second) for 01.. sort of molecules at the face of the cell, interpolation of ***b2stbr_rfluxm_eir001***